Annexure-E

GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY, ODISHA, GUNUPUR
Department of Computer Science & Engineering
Statement of Purpose – Major Project I

1. Title of the Project

Image Based Cattle Measurement and Trait Estimation System

2. Abstract

Animal Type Classification (ATC) plays a crucial role in dairy farming by assessing the body structure of cattle and buffaloes to predict productivity, longevity, and reproductive efficiency. Under national programs like the Rashtriya Gokul Mission (RGM), ATC is currently performed manually by trained personnel, making the process subjective, time-consuming, and prone to human error. This report proposes an AI-driven, image-based ATC system that automates the evaluation process using computer vision and deep learning techniques. By detecting anatomical landmarks from animal images, extracting body measurements, and converting them into standardized ATC scores, the proposed system ensures consistency, objectivity, and scalability.

3. Objectives

- To automate the Animal Type Classification process for cattle and buffaloes using AI and image processing.
- To reduce subjectivity, bias, and human error present in manual ATC methods.
- To accurately detect anatomical landmarks and extract key body structure parameters.
- To generate standardized and reproducible ATC scores.
- To enable seamless digital integration with the Bharat Pashudhan App.
- To support scientific decision-making in breeding programs under PT and PS schemes.
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4. Introduction / Problem Statement

Manual Animal Type Classification, despite being conducted by trained animal typers, is inherently subjective and susceptible to errors due to fatigue, visual bias, and measurement inaccuracies. These inconsistencies affect data quality and can lead to suboptimal selection of elite

dams under national breeding programs such as the Rashtriya Gokul Mission. With large-scale data collection required across diverse field conditions, there is a critical need for an automated, objective, and scalable system that can reliably evaluate animal body structure and assign standardized ATC scores with minimal human intervention.

5. Expected Input & Expected Output

Expected Inputs

- Digital images of cattle or buffaloes captured using mobile phones or cameras.
- Images may vary in background, lighting, and orientation under real field conditions.

Expected Outputs

- Detection of the animal and its key anatomical landmarks (e.g., shoulder, rump, hip, withers).
- Computed body structure measurements such as body length, rump angle, and body proportions.
- A standardized ATC score generated from normalized measurements.

6. Proposed Methodology / Techniques to be Used

- Image Acquisition : The user uploads or captures an image of a cattle or buffalo using a mobile or web interface.
- Animal Detection : A deep learning-based object detection model identifies and isolates the animal from the background.
- Landmark (Keypoint) Detection : A pose-estimation model detects anatomical landmarks critical for ATC evaluation.
- Measurement Extraction : Distances, angles, and ratios between landmarks are calculated to derive physical traits.
- Normalization & Scoring : Extracted measurements are normalized to remove scale and camera bias and converted into standardized ATC scores based on predefined criteria.
- Output Generation & Integration : Final scores and measurements are stored and prepared for integration with the Bharat Pashudhan App backend.

7. Novelty / Contribution

- Introduces a fully automated, image-based ATC system replacing subjective manual evaluation.
- Utilizes landmark-based scoring, ensuring consistency across evaluators and regions.

- Eliminates physical contact and manual measurement, improving animal welfare.
- Designed specifically for Indian field conditions.
- Enables objective, repeatable, and scalable evaluation, which is currently lacking in traditional ATC systems.

8. Dataset Description

The project uses a keypoint-annotated cattle image dataset consisting of RGB images captured in real farm environments with variations in breed, posture, lighting, and viewpoints. Each image is labeled with anatomical keypoints such as the shoulder, hip, rump, withers, and leg joints, enabling extraction of body measurements required for Animal Type Classification (ATC). The dataset is sourced from Roboflow Universe (Cow Keypoint Dataset) and was selected for its pre-annotated keypoints, real-world variability, compatibility with pose estimation models like YOLOv8-Pose, and public availability ensuring reproducibility.

9. Expected Outcomes / Deliverables

- A functional AI-based prototype for image-based Animal Type Classification.
- Accurate detection of animal body landmarks and extraction of physical traits.
- Standardized ATC scores generated with minimal human intervention.

10. Tools & Technologies

- Programming Language: Python
- Deep Learning Frameworks: PyTorch
- Computer Vision Models: YOLOv8, YOLOv8-Pose
- Image Processing: OpenCV, NumPy
- Backend & APIs: FastAPI
- Frontend (optional): React Native
- Model Training & Evaluation: Pandas, annotated datasets

Signature of the student **Signature of the supervisor**